Amber Handal

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EDUCATION

Northwestern University — Evanston, IL Sep 2025 – Dec 2026
M.S. in Robotics

University of Florida — Gainesville, FL May 2019 – May 2024
B.S. in Computer and Information Science Engineering

SKILLS

Reinforcement Learning	PPO, Actor-Critic, Reward Design, Domain Randomization, Sim-to-Real Transfer
Simulation & ML	Isaac Lab, Isaac Gym, MuJoCo, PyTorch, Deep Learning (CNNs), Computer Vision
Robotics	ROS/ROS 2, Visuomotor Policies, Contact Dynamics, Control, Kinematics, State Estimation, SLAM, Motion Planning, Humanoid & Legged Locomotion
Perception	RealSense RGB-D, Elevation Mapping, LiDAR, OpenCV, Sensor Calibration (Intrinsics, Extrinsics, Time Sync)
Software	Python, C++, C, TypeScript, Linux, Git, CI/CD, Docker, AWS

WORK EXPERIENCE

Reinforcement Learning Engineer Intern Persona AI — Pensacola, FL Jun 2026 – Present
- Established the company's first perceptive locomotion capability, leading development of vision-conditioned RL policies (PPO, Isaac Lab) for a Unitree G1 humanoid where only blind flat-ground policies existed.
- Train multiple perceptive policies against customer deployment requirements, building terrain curricula that mirror the client worksite: stairs, ramps and uneven ground, and cluttered narrow passages.
- Transfer trained policies from the G1 to the company's proprietary humanoid, retuning observations, action scaling, and rewards for its kinematics, actuator dynamics, and sensor placement.
- Architect a staged teacher-student pipeline – asymmetric privileged critic, symmetry augmentation, observation noise, and domain randomization – warm-started from a flat-walk policy to preserve gait priors.
- Own the observation contract end to end – a ray-cast height scan from RealSense D435i depth for terrain, and a torso-mounted AC1 LiDAR for state estimation – resolved into the robot's yaw-aligned waist frame.
- Lead hardware bring-up with the perception team: mounted and calibrated both sensors (intrinsics, extrinsics, time sync) and authored ROS 2 nodes streaming data to the on-robot policy at control rate.
- Validate sim-to-sim in MuJoCo before hardware, diagnosing a saturated height scan and control-rate mismatch.

Software Engineer Infotech — Gainesville, FL Sep 2022 – Aug 2025
- Built automated PDF pipelines using Python, TypeScript, and AWS Lambda to retrieve and classify plan data.
- Mentored three engineers to extend the system for DOT clients using token-based authentication and S3 APIs.
- Conducted code reviews and optimized Lambda concurrency and database queries for improved scalability.

PROJECTS

WatchDog — Autonomous Quadruped Inspection Jan 2026 – Mar 2026
- Built autonomous inspection system on Unitree Go2 with Nav2 and custom velocity command bridge.
- Configured RTAB-Map visual + ICP registration for SLAM using Unitree L2 LiDAR and RealSense D435i.
- Resolved multi-machine ROS 2 issues including DDS discovery, TF timestamp sync, and QoS tuning.
- Implemented frontier-based exploration for autonomous mapping with 2D LaserScan obstacle avoidance.
- Integrated SAM 3 on external GPU, projecting segmentation masks to 3D for object change detection.

EKF SLAM from Scratch — TurtleBot3 Localization & Mapping Jan 2026 – Mar 2026
- Engineered Extended Kalman Filter SLAM from scratch in C++ for TurtleBot3 using ROS 2 and Armadillo.
- Constructed EKF predict/correct cycle with Jacobian linearization for joint robot-landmark state estimation.
- Architected landmark detection with LiDAR clustering, SVD circle fitting, and inscribed angle classification.

PenPal — Vision-Guided Conversational Writing Robot Dec 2025
- Co-developed a ROS 2 system for a 7-DoF Franka arm to read and write on a moving whiteboard with 3 peers.
- Led perception using RealSense RGB-D, AprilTags, and OpenCV to estimate board pose and publish TF.
- Integrated a Gemini vision-language model for OCR + response generation via a ROS 2 service interface.
- Supported Cartesian writing execution by aligning the pen TCP to the board frame for consistent strokes.
- Generated font-based toolpaths by parsing OTF files to produce multi-language writing trajectories in 3D space.